

The double-dagger anisotropic norm is not equivalent in general

Solution packet for arXiv:math/0505015

11 August 2026

Candidate full negative answer. For every $1 < t < \infty$, $t \neq 2$, there are valid Baladi–Tsujii cone data in dimension three and $q < 0 < p$ for which the double-dagger norm is not equivalent to the anisotropic Sobolev norm on functions supported in any fixed compact set with nonempty interior. For $t = 2$ the norms are equivalent.

1 The question

Fix a compact set $K \subset \mathbb{R}^d$ with nonempty interior. A cone combination

$$\Theta = (C_+, C_-, \varphi_+, \varphi_-)$$

consists of closed cones with nonempty interiors and $C_+ \cap C_- = \{0\}$, together with smooth functions $\varphi_{\pm} : \mathbb{S}^{d-1} \rightarrow [0, 1]$ such that $\varphi_+ = 1$ on $C_+ \cap \mathbb{S}^{d-1}$, $\varphi_+ = 0$ on $C_- \cap \mathbb{S}^{d-1}$, and $\varphi_- = 1 - \varphi_+$. Put

$$P(\xi) = \langle \xi \rangle^p \varphi_+(\xi/|\xi|), \quad Q(\xi) = \langle \xi \rangle^q \varphi_-(\xi/|\xi|), \quad \langle \xi \rangle = (1 + |\xi|^2)^{1/2},$$

with harmless smooth definitions at $\xi = 0$. Appendix A of Baladi–Tsujii [1] introduces

$$N_{\dagger\dagger}(u) = \|(P(D) + Q(D))u\|_{L^t}, \quad N_{\dagger}(u) = \|P(D)u\|_{L^t} + \|Q(D)u\|_{L^t}.$$

It proves that $N_{\dagger}$ is equivalent to the paper’s anisotropic Sobolev norm and asks whether $N_{\dagger\dagger}$ is equivalent to that norm. The triangle inequality gives $N_{\dagger\dagger} \leq N_{\dagger}$; the issue is the reverse inequality.

2 Statement of the answer

Theorem 1. *Let $K \subset \mathbb{R}^3$ be compact with nonempty interior and let $1 < t < \infty$, $t \neq 2$. There are a cone combination Θ and real numbers $q < 0 < p$, with $p - q$ arbitrarily small, for which no constant C satisfies*

$$\|P(D)u\|_{L^t} + \|Q(D)u\|_{L^t} \leq C\|(P(D) + Q(D))u\|_{L^t}, \quad u \in C_c^\infty(K). \quad (1)$$

Consequently $N_{\dagger\dagger}$ is not equivalent in general to the Baladi–Tsujii anisotropic Sobolev norm.

Remark 2. *For $t = 2$ equivalence always holds. The symbols P, Q are nonnegative, so Plancherel and $P^2 + Q^2 \leq (P + Q)^2$ give*

$$N_{\dagger}(u) \leq \sqrt{2}(\|P(D)u\|_2^2 + \|Q(D)u\|_2^2)^{1/2} \leq \sqrt{2} N_{\dagger\dagger}(u).$$

Let $\|\cdot\|_{W^{\Theta,p,q,t}}$ be the norm defined in Section 2. We shall prove below (using Theorem 3.1) that for each $1 < t < \infty$ there is a constant $C > 0$ so that

$$(37) \quad C^{-1}\|u\|_{W_*^{\Theta,p,q,t}} \leq \|u\|_{W_*^{\Theta,p,q,t}}^{\dagger} \leq C\|u\|_{W_*^{\Theta,p,q,t}} \quad \text{for } u \in C^\infty(K).$$

Let $W_{\dagger}^{\Theta,p,q,t}(K)$ be the completion of $C^\infty(K)$ with respect to $\|\cdot\|_{W_*^{\Theta,p,q,t}}^{\dagger}$. Then $W_{\dagger}^{\Theta,p,q,t}(K) = W_*^{\Theta,p,q,t}(K)$ by (37).

Let $W_{\dagger\dagger}^{\Theta,p,q,t}(K)$ be the completion of $C^\infty(K)$ with respect to $\|\cdot\|_{W_*^{\Theta,p,q,t}}^{\dagger\dagger}$. Then

$$(38) \quad W_{\dagger\dagger}^{\Theta,p,q,t}(K) = \{u \in \mathcal{S}'(\mathbb{R}^d) \mid \text{supp}(u) \subset K, \quad \|u\|_{W_*^{\Theta,p,q,t}}^{\dagger\dagger} < \infty\}, \forall 1 < t < \infty.$$

Clearly,

$$(39) \quad \|u\|_{W_*^{\Theta,p,q,t}}^{\dagger\dagger} \leq \|u\|_{W_*^{\Theta,p,q,t}}^{\dagger} \quad \text{for } u \in C^\infty(K).$$

Though we do not know whether the norm $\|\cdot\|_{W_*^{\Theta,p,q,t}}^{\dagger\dagger}$ is equivalent to $\|\cdot\|_{W_*^{\Theta,p,q,t}}$, we show below (using Theorem 3.1) that, for each $1 < t < \infty$, if $\Theta' > \Theta$ and $p \geq q$,

$$(40) \quad \|u\|_{W_*^{\Theta,p,q,t}}^{\dagger} \leq C\|u\|_{W_*^{\Theta',p,q,t}}^{\dagger\dagger} \quad \text{for } u \in C^\infty(K)$$

for some constant $C > 0$. From (37), (39) and (40), it follows that, if $\Theta' > \Theta$ then for all $1 < t < \infty$ and $p \geq q$, then $W_*^{\Theta',p,q,t}(K) \subset W_*^{\Theta,p,q,t}(K)$, and, also,

$$W_{\dagger\dagger}^{\Theta',p,q,t}(K) \subset W_*^{\Theta,p,q,t}(K) \subset W_{\dagger\dagger}^{\Theta,p,q,t}(K).$$

Let $W_{\dagger\dagger}^{\Theta,p,q,t}(\mathbb{R}^d) \subset \mathcal{S}'(\mathbb{R}^d)$ be the isometric image of the restriction to $L^t(\mathbb{R}^d)$ of the inverse of the bijective operator $\Psi_{\Theta,p,+}(D) + \Psi_{\Theta,q,-}(D) : \mathcal{S}'(\mathbb{R}^d) \rightarrow \mathcal{S}'(\mathbb{R}^d)$. For $1 < t < \infty$, (38) implies that $W_{\dagger\dagger}^{\Theta,p,q,t}(K)$ is just (isometrically) $\{u \in W_{\dagger\dagger}^{\Theta,p,q,t}(\mathbb{R}^d) \mid \text{supp}(u) \subset K\}$. The space $W_{\dagger}^{\Theta,p,q,t}(K)$ may be described in a similar (although not as neat) way using the injective (non surjective) operator $(\Psi_{\Theta,p,+}(D), \Psi_{\Theta,q,-}(D)) :$

Figure 1: The exact norm-equivalence question on page 19 of the official arXiv PDF.

3 The cone and its limiting projection

Fix $0 < c < 1$ and take the circular cones

$$C_- = \{\xi \in \mathbb{R}^3 : \xi_3 \geq c|\xi|\}, \quad C_+ = \{\xi \in \mathbb{R}^3 : \xi_3 \leq -c|\xi|\}.$$

Choose $\varphi_+ \in C^\infty(\mathbb{S}^2; [0, 1])$ which is zero exactly on $C_- \cap \mathbb{S}^2$, is one on $C_+ \cap \mathbb{S}^2$, and is strictly between zero and one elsewhere; standard flat cutoffs give such a function. Set $\varphi_- = 1 - \varphi_+$. Choose a small $s > 0$ and put

$$p = s/2, \quad q = -s/2.$$

Let $S = P + Q$ and define, for $R \geq 1$,

$$m_R(\xi) = \frac{P(R\xi)}{S(R\xi)}.$$

For every $\xi \neq 0$ away from the boundary of C_- ,

$$m_R(\xi) \longrightarrow m_\infty(\xi) := \mathbf{1}_{\mathbb{R}^3 \setminus C_-}(\xi). \quad (2)$$

Indeed, off C_- the positive φ_+ term has order $R^{s/2}$ and dominates the $R^{-s/2}$ term, while on C_- the numerator vanishes.

The slice of C_- by $\xi_3 = 1$ is the disk

$$\{\xi' \in \mathbb{R}^2 : |\xi'| \leq (c^{-2} - 1)^{1/2}\}.$$

Thus the affine slice of m_∞ is the characteristic function of the exterior of a disk. Fefferman's ball-multiplier theorem [3], together with de Leeuw restriction [2], says that this symbol is not even a local $L^t(\mathbb{R}^2)$ multiplier near its circular boundary unless $t = 2$.

4 Keeping the support inside K

We give the localization step which prevents the fixed support condition from hiding the multiplier obstruction.

Lemma 3 (expanding-support quotient limit). *For the cone data above, suppose that for some C ,*

$$\|P(D)u\|_{L^t} \leq C\|S(D)u\|_{L^t}, \quad u \in C_c^\infty(K). \quad (3)$$

If $s > 0$ is sufficiently small, then m_∞ is a local L^t multiplier on every frequency annulus.

Proof. Choose x_0 and $r_0 > 0$ with $B(x_0, r_0) \subset \text{int } K$. Changing variables $y = R(x - x_0)$ in (3) gives

$$\|P(RD)v\|_{L^t} \leq C\|S(RD)v\|_{L^t} \quad (4)$$

whenever $\text{supp } v \subset B(0, r_0 R)$.

Let U be a compact subset of an annulus and take a Schwartz function w with $\widehat{w} \in C_c^\infty(U)$. Define

$$v_R = S(RD)^{-1}w.$$

This is Schwartz for every fixed R . On U , the denominator in $S(R\xi)$ is bounded below, after factoring out $R^{s/2}$, by a constant times R^{-s} . Repeated differentiation of the reciprocal therefore gives

$$\left| \partial_\xi^\alpha \frac{1}{S(R\xi)} \right| \leq C_{\alpha,U} R^{s(|\alpha|+1)}. \quad (5)$$

The powers in this deliberately loose estimate are all that is needed.

Choose numbers $s < \beta < 1$ and a cutoff $\chi_R(x) = \chi(x/R^\beta)$, where $\chi = 1$ on the unit ball and is supported in the ball of radius two. Fourier integration by parts using (5) shows, for every fixed Sobolev order a and every N ,

$$\|(1 - \chi_R)v_R\|_{W^{a,t}} \leq C_{a,N,U} R^{s(N+C_a)-\beta N}.$$

Taking N large (and initially taking s smaller if necessary) gives

$$\|S(RD)(\chi_R v_R) - w\|_{L^t} = o(1), \quad (6)$$

$$\|P(RD)(\chi_R v_R) - m_R(D)w\|_{L^t} = o(1). \quad (7)$$

Here one uses the standard Mihlin bounds $\|P(RD)f\|_t + \|S(RD)f\|_t \leq CR^{s/2}\|f\|_{W^{s/2,t}}$ for the tails. Since $R^\beta = o(R)$, the function $\chi_R v_R$ is admissible in (4) for all large R . Equations (4)–(7) yield

$$\|m_R(D)w\|_{L^t} \leq C\|w\|_{L^t} + o(1). \quad (8)$$

The symbols satisfy $0 \leq m_R \leq 1$ and converge almost everywhere on U to m_∞ . Hence $m_R(D)w \rightarrow m_\infty(D)w$ in L^2 . Taking an almost-everywhere convergent subsequence and applying Fatou to (8) gives

$$\|m_\infty(D)w\|_{L^t} \leq C\|w\|_{L^t}.$$

This is the asserted local multiplier estimate. $\square$

5 Proof of the counterexample

Proof of Theorem 1. If (1) held, then its first summand would give (3). Lemma 3 would make the characteristic function of the complement of the circular cone a local $L^t(\mathbb{R}^3)$ multiplier.

Translate frequency by $(0, 0, 1)$ and apply the local affine form of de Leeuw's restriction theorem to the plane $\xi_3 = 0$. The restricted symbol is the characteristic function of the exterior of the disk $|\xi'| \leq (c^{-2} - 1)^{1/2}$. Fefferman's theorem says that a disk characteristic is a local $L^t(\mathbb{R}^2)$ multiplier only for $t = 2$. This contradicts the chosen $t \neq 2$, proving that (1) fails.

The parameter $s = p - q$ was required only to be positive and sufficiently small, so the gap can be arbitrarily small while retaining $q < 0 < p$. Finally, the equivalence between $N_\dagger$ and the original anisotropic Sobolev norm is equation (37) of the source, so failure of (1) is exactly failure of the equivalence asked there. $\square$

6 Novelty and review

The run indexes, parsed corpus, exact source phrase, published record, and bounded primary-source searches through 11 August 2026 found no later answer to this question. The argument combines classical multiplier obstructions in a way not located in the anisotropic-space literature. Novelty confidence is moderate.

The proof is analytic and uses no numerical evidence. Expert review should focus on the graph-norm cutoff in Lemma 3 and on the standard local affine formulation of de Leeuw restriction. The theorem gives a complete negative answer to equivalence in general; it does not claim failure for every cone combination.

References

- [1] V. Baladi and M. Tsujii, *Anisotropic Hölder and Sobolev spaces for hyperbolic diffeomorphisms*, Ann. Inst. Fourier **57** (2007), 127–154; arXiv:math/0505015.
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- [3] C. Fefferman, *The multiplier problem for the ball*, Ann. of Math. **94** (1971), 330–336.